

ADVANCED PROGRAMMING PRACTICE (21CSC203P)

Unit – 1 : Session – 3 : SLO – 1 | Bohm-Jacopini Theorem

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Activity (Solutions)

- 1) Three control structures in Java — SEQUENCE: `int a = 5; int b = 10; int c = a + b;` (statements run in order). SELECTION: `if (c > 10) System.out.println("big"); else System.out.println("small");` ITERATION: `for (int i = 0; i < 3; i++) System.out.println(i);` — Bohm-Jacopini theorem: any algorithm can be expressed using only these three structures.
- 2) goto-like (nested if-break) converted to structured code — Unstructured: `outer: while(true){ read(x); if(x<0) break outer; if(x==0) continue; process(x); }` Structured: `int x = read(); while (x >= 0) { if (x != 0) process(x); x = read(); }` — same logic using a loop condition instead of break-to-label.
- 3) Importance of structure: single-entry/single-exit blocks make control flow predictable, so code is easier to read, debug, test and maintain; it eliminates spaghetti code caused by arbitrary jumps, enables modular reasoning, and lets compilers optimize better.